Brier: Confidence-Calibrated Slashing for On-Chain AI Decision Accountability

A research proposal with a measured v0 prototype

Version v0 (prototype)
Date August 2026
Repository github.com/Sagexd08/Brier

This is a mechanism and a measured prototype, not a deployable protocol.
Only the calibration head is zero-knowledge proved — its input logit is unverified, dispute resolution rests on an admin-appointed N-of-M committee whose collusion carries the authority a single key would, and the figures reported here come from a

local chain. Section 7 states each limitation in full; Section 8 separates what has shipped from what remains.

Contents

Abstract

Accountability mechanisms for automated decisions penalise error at a confidence-independent rate, so an operator is never punished for asserting 99% confidence in a decision it believes at 60%. We propose slashing staked collateral by the **Brier score** over (reported confidence, adjudicated outcome). Because that score is strictly proper, expected loss is minimised exactly when the operator reports its true subjective probability, making honest confidence reporting loss-minimising rather than good faith. We report a measured prototype: across 10 pinned seeds, temperature scaling cuts Expected Calibration Error from 0.1853 ± 0.0248 to 0.0870 ± 0.0218 on held-out UCI German Credit data, in every seed; an EZKL/halo2 circuit proves the calibration head in 2.09 s, verified on-chain for 684,696 gas. This is a mechanism and a prototype, **not a deployable protocol**: only the calibration head is proved and its input logit is unverified, and dispute resolution rests on an admin-appointed N-of-M committee whose collusion has exactly the power a single key had. An unbonding period now closes the front-running exit that broke v0's economic tier, leaving two residual gaps that a longer timelock cannot fix. The proposal states what each remaining gap would take to close, and what evidence would settle whether the mechanism survives contact with them. A subsequent pass ablated five candidate enhancements against this baseline: conformal prediction earns a place and fits the circuit budget at no measurable cost, while a deeper base model and deep-ensemble epistemic uncertainty do not — the latter measurably *harms* calibration. Both negative results are reported at the same prominence as the positive ones.

1. Introduction

1.1 The regulatory setting

Automated credit decisions are moving into a compliance regime that demands records rather than assurances.

Under the EU AI Act, credit scoring is a high-risk application, and Article 26 places operational duties on **deployers** of such systems. Article 26(5) requires deployers to "monitor the operation of the high-risk AI system on the basis of the instructions for use" and to notify providers of risks or serious incidents without undue delay. Article 26(6) requires deployers to "keep the logs automatically generated by that high-risk AI system … for a period appropriate to the intended purpose … of at least six months." Article 86 establishes a right to explanation of individual decision-making. Article 26 becomes applicable on 2 December 2027 for most high-risk systems, with an extended deadline of 2 August 2028 for systems in employment, education, and access to essential services [1].

These provisions mandate *logging and explanation*. They do not establish that the logs are truthful, nor do they price the difference between a wrong decision made tentatively and a wrong decision made with asserted certainty. A deployer that logs a confidently wrong decision is compliant with Article 26(6).

A note on the Indian setting, correcting a common assumption. The RBI Digital Lending Directions, 2025 are frequently cited as imposing algorithmic transparency duties. They do not. The RBI Working Group recommended that regulated entities "document the rationale for algorithmic features and conduct regular algorithmic audits," but these recommendations were incorporated into neither the 2022 Guidelines nor the 2025 Directions, which remain silent on algorithmic fairness and transparency [2]. We therefore cite India as an example of a **regulatory gap** rather than a regulatory driver. We were unable to locate a provision in the 2025 Directions imposing model-explainability obligations, and we do not assert one exists.

The honest framing is that regulation currently creates demand for *auditable records of automated decisions* and does not yet create demand for *calibrated confidence*. This proposal argues the latter is the more useful object, not that it is presently required.

1.2 The gap in existing mechanisms

Decentralised protocols already slash staked collateral for misbehaviour, and already run decentralised claim adjudication. Neither weights the penalty by the reporter's own stated confidence.

- **Oracle slashing.** Chainlink slashes node operators for failing on-chain service-level obligations at a **fixed 700 LINK per valid alerting event** on the ETH/USD data feed, with the alerter rewarded 7,000 LINK [3]. The penalty is a constant: it is identical whether the operator was marginally or egregiously wrong, and oracles do not report confidence at all.
- **Decentralised insurance.** Nexus Mutual resolves claims by stake-weighted member voting, with claim assessors staking NXM and rewarded for voting with the outcome [4]. The mechanism adjudicates *whether a claim is valid*; it does not score how confident anyone was.
- **On-chain proper scoring rules.** These do exist. Foresight Arena (arXiv:2605.00420) scores AI forecasting agents on Polymarket questions using the Brier score, with commitments and scores recorded on Polygon PoS via smart contracts [5].

The precise delta is therefore narrower than "nothing like this exists," and we state it as such: **proper scoring rules have been applied on-chain to *reward* forecasters in prediction-market settings; we are not aware of prior work applying one to *slash staked collateral* of a model operator under dispute, with the scored quantity produced by a zero-knowledge-proved calibration step.** Foresight Arena is reward-based, involves no staked collateral at risk, no slashing, and no zero-knowledge proofs [5]. We claim novelty in the combination, not in any component.

1.3 Research questions

The proposal is organised around five questions. Each names the evidence that would answer it; §6 reports how far the v0 prototype gets, and §8 proposes the work that would close the remainder.

RQ1 — Incentive. Does penalising a disputed decision by its Brier score make truthful confidence reporting the operator's loss-minimising strategy, and does that property survive implementation in fixed-point on-chain arithmetic? *Evidence:* an analytic derivation (§3.2), plus numerical verification of the deployed Solidity against it (§5.6, §6.11).

RQ2 — Feasibility. Can the confidence-producing step be proved in zero knowledge and verified on-chain at a cost a per-decision workflow could absorb? *Evidence:* measured proving time, proof size, and verification gas (§6.3, §6.4).

RQ3 — Cost structure. How does proving cost scale with the size of the proved head, and does that scaling favour proving a small calibration head over a full classifier? *Evidence:* a parameter sweep across four orders of magnitude (§6.3).

RQ4 — Sufficiency. Is a proved calibration head enough to make the resulting attestation trustworthy? *Evidence:* an adversarial reading of the trust boundary, with each claimed weakness executed as a test (§4.3, §7). The answer established here is **no**, and that answer is what §8 is addressed to.

RQ5 — Sophistication. Does a more capable model, or a stronger uncertainty quantifier, improve the quantity this mechanism actually prices? *Evidence:* five enhancements ablated against the shipped baseline on the identical seed protocol, each kept only if it earns its place (§6.5–§6.9, and `ABLATION.md`).

1.4 Contributions

1. A slashing rule built on a strictly proper scoring rule, with the properness argument derived in closed form (§3.2) and its scope conditions stated as part of the claim rather than as trailing caveats (§3.3).
2. A working implementation: fixed-point Brier arithmetic in Solidity at 543 gas, an EZKL/halo2 circuit over the calibration head, and an attestation contract that rejects any decision whose proof does not verify.
3. A measured evaluation over 10 pinned seeds, including a leakage control that comes out *worse than not calibrating at all*, and a negative result on head capacity reported at the strength the data supports rather than the strength that would read better.
4. An adversarial trust model (§4.3) in which every asserted weakness is demonstrated by an executing test, so the threat model cannot drift away from the code without the test suite failing.
5. A conformal-prediction layer carrying a distribution-free coverage guarantee, shown by measurement to fit the existing circuit budget — and an ablation of four further enhancements, two of which are reported as failures because they are (§6.5–§6.9).
6. Two on-chain surfaces — an operator reputation register and a content-addressed model-version registry — each placed explicitly in the trust tier it actually occupies rather than the one its being on-chain might suggest (§4.3, §6.10).

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2. Related work

2.1 zkML and verifiable inference

EZKL compiles computational graphs supplied in ONNX format into arithmetic circuits and produces zk-SNARK proofs of correct model execution, using a modified halo2 with Plonkish arithmetisation; anyone holding the verifying key can check the proof [6]. This prototype uses EZKL 23.0.5 directly. The broader design question EZKL raises — that proving cost scales with the proved subgraph — is what motivates proving a small calibration head rather than a full classifier, and is examined empirically in §6.3.

2.2 Decentralised oracle staking and slashing

Chainlink's staking design uses on-chain service-level agreements, community alerting, and fixed-size slashing penalties [3]. Nexus Mutual's claim assessment uses stake-weighted voting with a quorum, escalation to a full member vote, and reward for majority-aligned voting [4]. Kleros provides general decentralised arbitration: jurors are drawn from PNK stakers, self-select into sub-courts, vote on evidence, and are rewarded for voting with the majority under a Schelling-point incentive, with appeals heard by successively larger juries [7]. Kleros is the most plausible substitute for this prototype's trusted administrator (§8).

2.3 Calibration in machine learning

Guo et al. show that modern neural networks are poorly calibrated, and that temperature scaling — a single-parameter extension of Platt scaling that divides logits by a learned $T > 0$ fitted on held-out data — is surprisingly effective at restoring calibration [8]. Temperature scaling is the primary calibration method here, chosen both for its empirical strength and because a single-parameter map is trivially cheap to arithmetise.

2.4 Proper scoring rules

Brier introduced the mean-squared-error score for probability forecasts of binary outcomes [9]. Gneiting and Raftery develop the general theory: a scoring rule is *proper* if the forecaster maximises expected score by issuing its true predictive distribution F rather than any $G \neq F$, and *strictly proper* if that maximum is unique [10]. The Brier score is strictly proper, which is the sole property this mechanism depends on.

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3. The mechanism

3.1 The slashing rule

Let S be the collateral an operator has staked, $c \in [0,1]$ the confidence it reported that its decision was correct, and $o \in \{0,1\}$ the adjudicated outcome of a dispute over that decision. On an upheld dispute the contract slashes

$$slash = S \cdot (c - o)^2$$

subject to a protocol maximum fraction of S. The penalty is the Brier score of a single forecast, denominated in stake.

3.2 Why truthful reporting minimises expected loss

Suppose the operator privately believes the decision is correct with probability p, and that a dispute, if opened, is adjudicated truthfully — so $o = 1$ with probability p. Reporting c, its expected slash is

$$E[slash] = S\,(\,p(c\text{-}1)^2 + (1\text{-}p)c^2\,) = S\,(\,c^2 - 2pc + p\,)$$

and completing the square in c gives

$$E[slash] = S\,(\,(c\text{-}p)^2 + p(1\text{-}p)\,)$$

Two things follow, and together they are the whole argument.

The unique minimum is at $c = p$. The expression is a strictly convex parabola in c, so any misreport in either direction costs the operator $S(c\text{-}p)^2$ in expectation. Overclaiming confidence is not merely unrewarded, it is priced — and priced quadratically, so shading the truth slightly is cheap while asserting near-certainty about a coin flip costs close to the entire stake. This is strict properness [10] instantiated as a collateral rule: the operator does not need to be honest, it needs only to be selfish.

The residual $S \cdot p(1\text{-}p)$ is irreducible. It does not depend on what the operator reports. An operator facing genuine uncertainty pays for that uncertainty however honestly it reports, which is the economically correct reading: the term is the price of writing an opinion on a hard case, and it is maximised at $p = 0.5$, where the model knows least. The mechanism cleanly separates the cost of *being uncertain* from the cost of *misrepresenting uncertainty*, and only the second is avoidable by the operator.

The deployed contract computes this in fixed point rather than over the reals, so properness is re-established empirically against the shipped code rather than inherited from the algebra: §5.6 scans all 101 candidate reports at two distinct true probabilities and confirms the expected-loss minimiser is the true one in both, and monotonicity is fuzzed at 256 runs per direction.

3.3 Scope conditions, stated as part of the claim

The derivation assumes the adjudicated outcome is truthful and that dispute selection is independent of the reported confidence. Neither condition is guaranteed by this prototype. Specifically:

1. **Input-logit provenance is unproven.** The circuit takes the base model's logit as an unverified public input. An operator that fabricates the logit obtains a proof that verifies. The proof binds the calibration step, not the pipeline (Figure 3, tier 1; `ThreatModel.t.sol::test_tier1_marginIsUnverifiedOperatorSuppliedInput`).
2. **Off-chain misreporting before proving is not prevented.** Nothing forces the operator to feed the circuit the logit its deployed model actually produced.
3. **Selective disputes break the properness argument.** If only rejections are ever disputed, the operator faces an asymmetric loss surface, and the dominant strategy under that surface is not analysed here.

The mechanism is proper; the *system* is proper only to the extent that these conditions hold, and in v0 they do not. Closing conditions 1 and 3 is the substance of the proposed work in §8.

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4. System design

4.1 Components and where the circuit sits

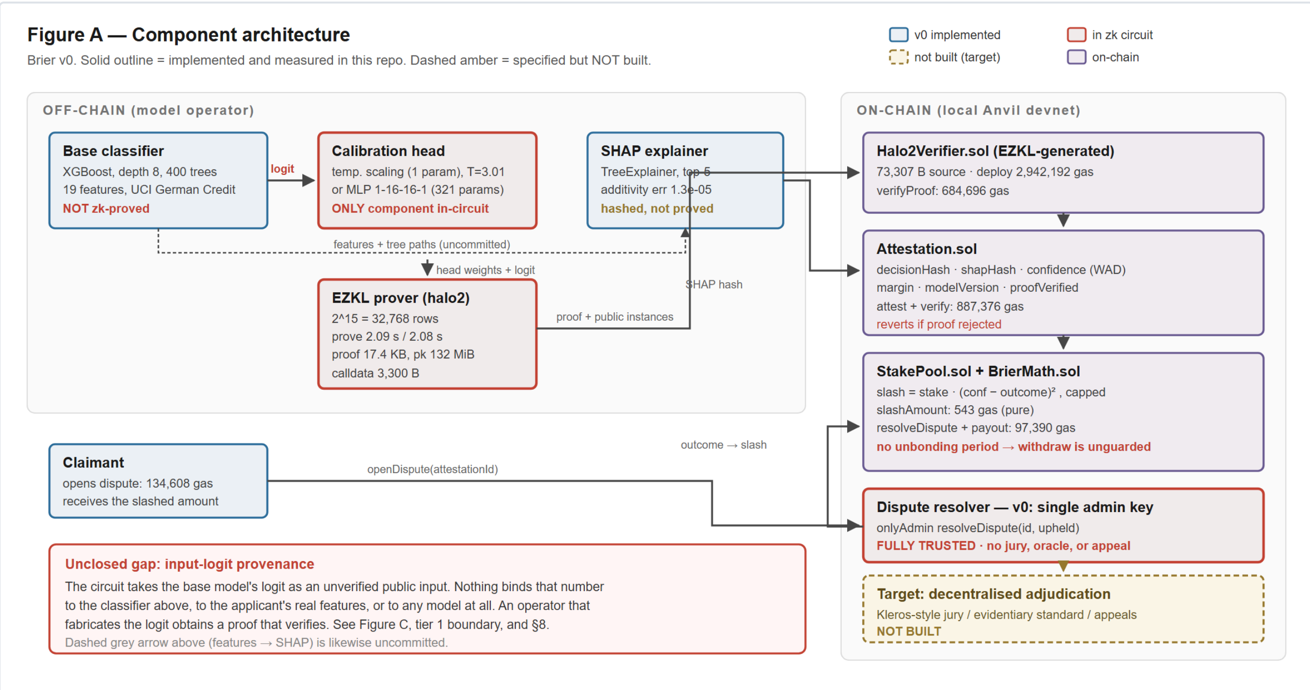

Figure 1 — Component architecture. Solid outlines are implemented and measured; dashed amber is specified but unbuilt; red is the single component inside the zk circuit.

Solid outlines mark components implemented and measured in the repository; dashed amber marks specified-but-unbuilt. Red marks the single component inside the zk circuit. The base classifier and the SHAP explainer sit outside the circuit by design; the SHAP vector is hash-committed as evidence only.

4.2 Protocol sequence and per-step cost

The sequence below is annotated with the gas each step actually cost on a local chain, so the expensive steps are visible rather than asserted: verification dominates, and the Brier arithmetic that is the substance of the mechanism is three orders of magnitude cheaper than proving that it ran. Two notes in the diagram mark where v0 computes a penalty it cannot enforce — the operator's unrestricted `withdraw()` between attestation and dispute (§7.3), and the single administrative key that decides the outcome (§7.2).

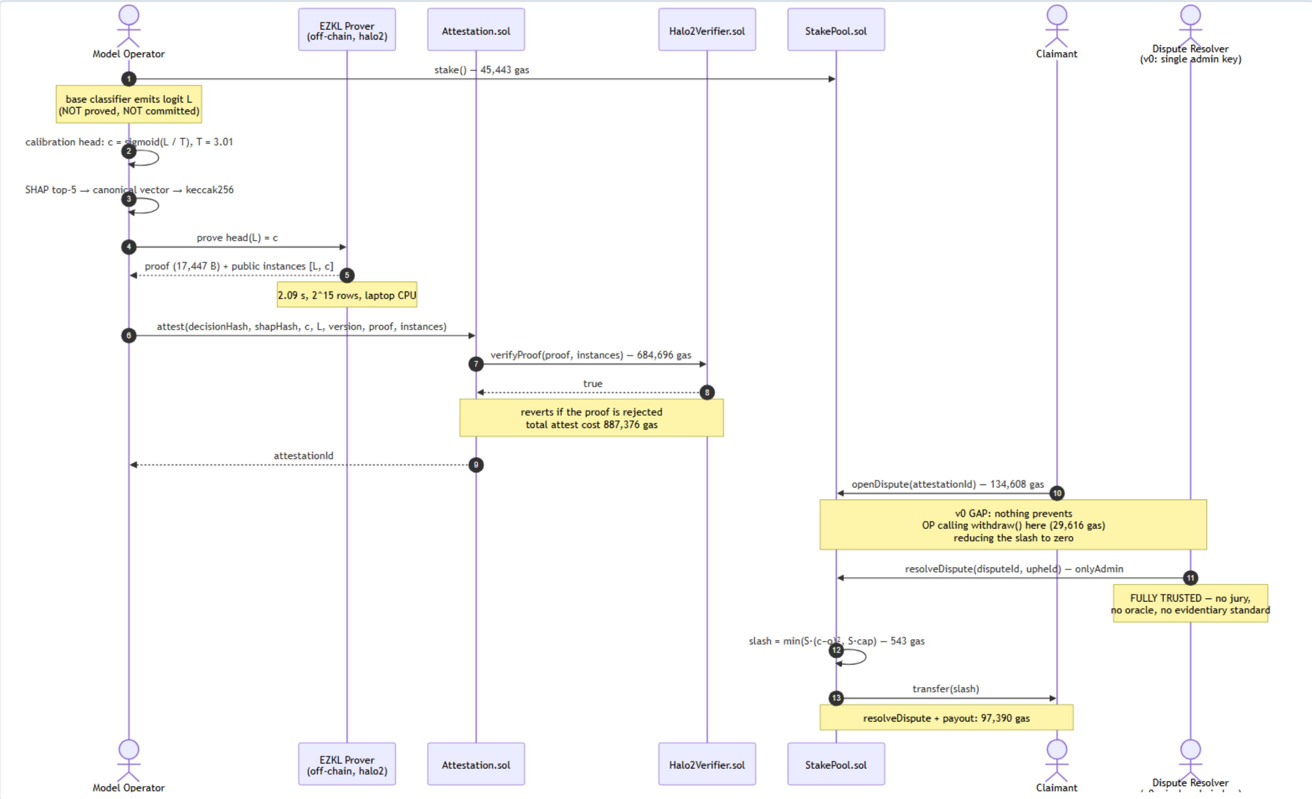

Figure 2 — Protocol sequence, annotated with measured gas per step and the two points at which v0 is unenforced.

4.3 Trust boundaries

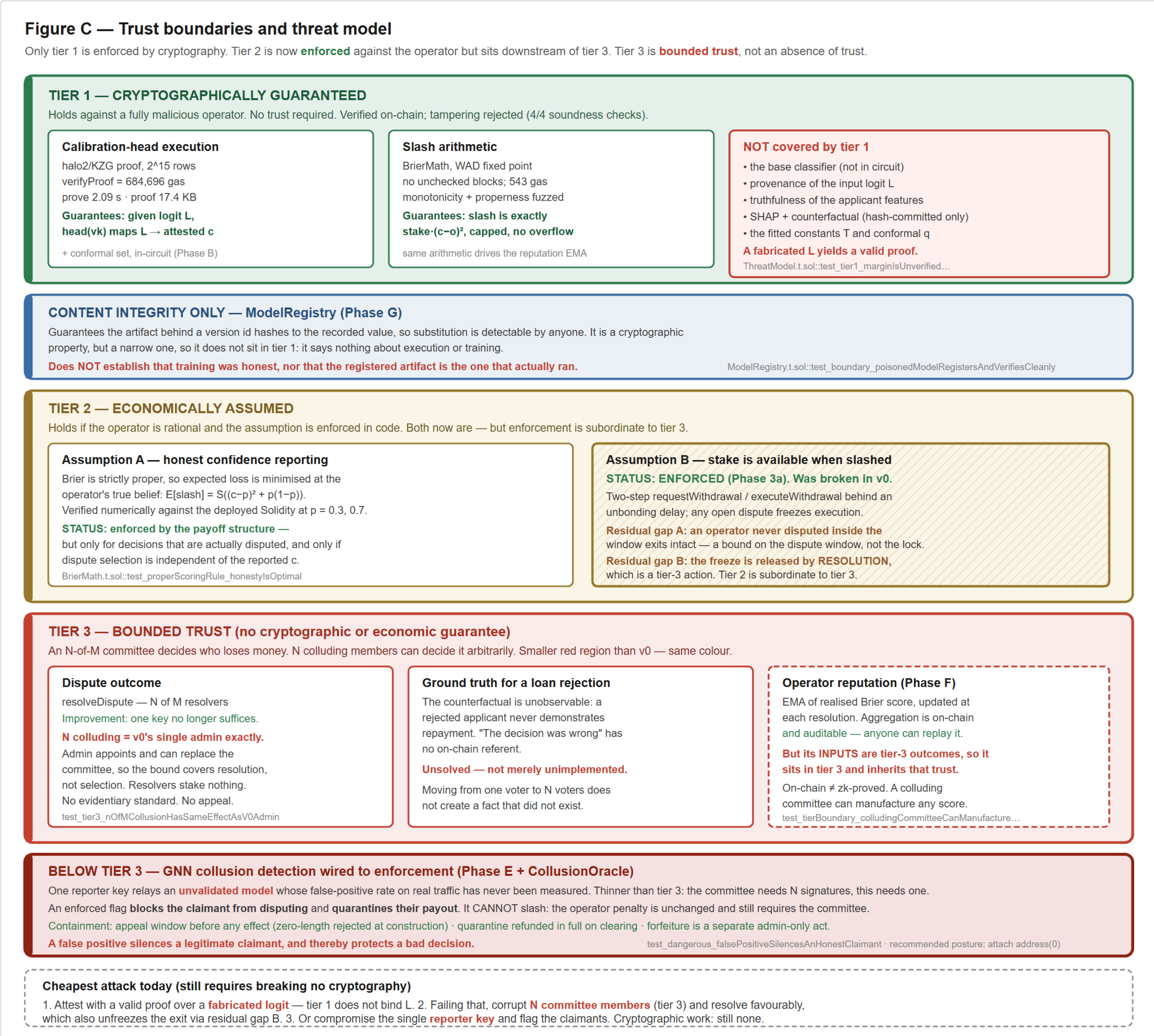

Figure 3 — Trust boundaries. Green is cryptographically guaranteed, amber economically assumed, red fully trusted; each weakness names the test that demonstrates it.

This is the diagram against which every other claim in this document should be checked. Three tiers:

- **Tier 1 (cryptographically guaranteed, green).** Calibration-head execution and the slash arithmetic. Holds against a fully malicious operator.
- **Tier 2 (economically assumed, amber).** Assumption A (honest confidence reporting) is enforced by the payoff structure, conditionally. Assumption B (stake is available when slashed) was **broken in v0** and is now enforced: withdrawal is a two-step request/execute with an unbonding delay, and any open dispute freezes execution (§7.3). Two residual gaps remain, and neither is a timelock bug.
- **Tier 3 (bounded trust, red).** Dispute outcome and ground truth. An admin-appointed N-of-M committee decides who loses money; N colluding members can decide it arbitrarily. The tier keeps its colour — it is a smaller red region, not a different one.

Each weakness in the figure names the test that demonstrates it. The cheapest attack path still requires no cryptographic work: fabricate the logit, since tier 1 does not bind it (`test_tier1_marginIsUnverifiedOperatorSuppliedInput`), or corrupt N committee members, which `test_tier3_nOfMCollusionHasSameEffectAsV0Admin` shows buys exactly the power v0's single key had.

4.4 Calibration reliability

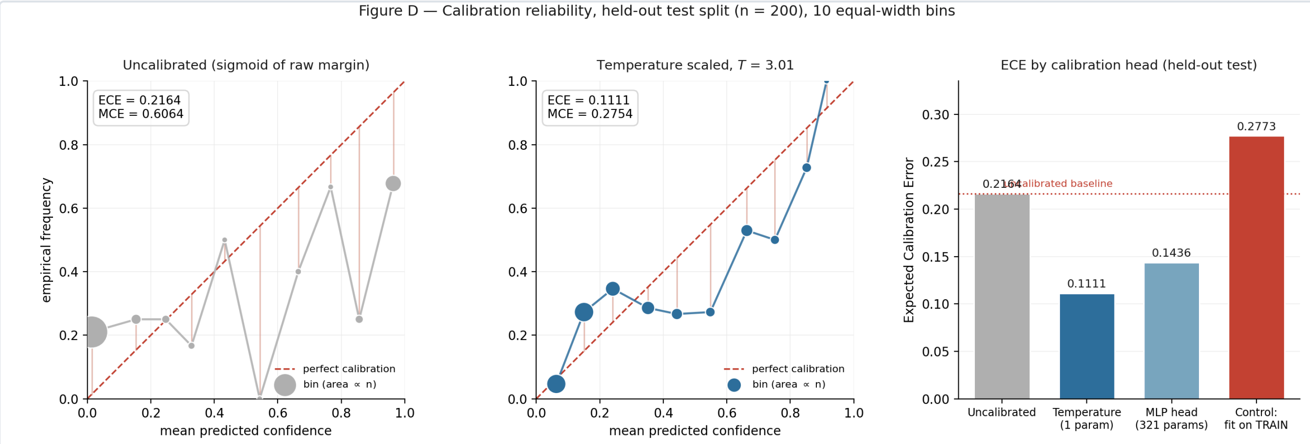

Figure 4 — Reliability before and after temperature scaling, with the train-fitted leakage control at right. Bin values are read directly from the run artifacts; no curve is fitted.

Bin values are read directly from `artifacts/calibration/phase1_report.json`; no curve is fitted or smoothed. Left: uncalibrated, ECE 0.2164, with a bin at mean confidence 0.5430 whose empirical frequency is 0.0000. Centre: after temperature scaling with $T = 3.01$, ECE 0.1111. Right: ECE across heads, including the leakage control described in §5.2.

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5. Methodology

5.1 Dataset and model

UCI Statlog German Credit: 1,000 real credit applications, 20 attributes, downloaded at build time. The label is inverted so that the positive class is *reject*. Attribute 9 (personal status and sex) is dropped, leaving 19 features (see §7.4). Split 60/20/20 into train (600) / calibration (200) / test (200), stratified, seed 42, with mutual disjointness asserted in code.

The base model is XGBoost (depth 8, 400 trees, learning rate 0.3, no regularisation) and is **deliberately overfit**: train accuracy 1.0000, test accuracy 0.7150. This is a methodological choice, not a tuned result. Demonstrating a calibration-insurance mechanism on an already-calibrated model would establish nothing; the overfit regime reproduces the overconfidence that makes calibration worth insuring. The base model's accuracy is not a claim of this work.

5.2 Calibration

Two heads were implemented and both are proved:

Head	Parameters	Form
Temperature scaling	1	$\sigma(L/T)$, T learned in log-space
MLP	321	$1 \to 16 \to 16 \to 1$, ReLU

Both emit a **logit**, with the sigmoid applied outside the module, so the circuit is affine + ReLU only and contains no transcendental function.

Both are fitted on the **calibration split only**. To show this matters rather than assert it, every run includes a control that fits the same head on the *training* split: that control yields $T = 0.1344$ and test ECE 0.2773, which is **worse than not calibrating at all** (0.2164). Fitting on training data *sharpens* an already-overconfident model.

5.3 Uncertainty quantification beyond a point estimate

Temperature scaling returns a calibrated scalar with no guarantee attached. Two stronger constructions were implemented and evaluated against it.

Split conformal prediction returns a *set* carrying a distribution-free marginal coverage guarantee: over the joint draw of calibration and test data, the true label lies in the set at least $1\text{-}alpha$ of the time, with no assumption that the model is correct or even well specified. The calibration split is halved so the temperature and the conformal quantile never see the same points — reusing one split for both would break the exchangeability the guarantee rests on. Results in §6.5.

Deep-ensemble epistemic uncertainty trains N independently seeded copies of the neural base model and feeds the spread of their predictions to the calibration head alongside the margin, on the intuition that a confident score the members disagree about should be discounted. Results in §6.6, where that intuition does not survive measurement.

5.4 Explainability

SHAP TreeExplainer over the base model, top-5 attributions per decision, in margin space. Correctness is checked by additivity (SHAP values plus base value must reconstruct the model margin; measured max absolute error 1.26×10^{-5}) and by bit-identical reproduction across reruns. Five directional claims, written before inspecting output, are verified by correlating each feature's value against its attribution toward reject.

Counterfactual explanations. SHAP answers "what drove this decision"; an applicant reading an adverse action notice is asking "what would change it". Those are different questions, and the second is the actionable one. A greedy coordinate search over *actionable* features produces a minimal-ish change set that flips a rejection, restricted to values the training data actually contains. Immutable attributes — age, foreign-worker status, dependants — are excluded by construction and the exclusion is asserted rather than assumed. The counterfactual is hash-committed alongside the SHAP vector, in the same evidence slot and at the same trust tier: it is proof the operator recorded this explanation, not proof the explanation is faithful. Results in §6.7.

5.5 Evaluation metrics

ECE (10 equal-width bins, weighted by bin population, empty bins skipped rather than counted as perfectly calibrated), MCE, Brier score, proof generation time, proof size, and on-chain verification gas. The ECE implementation is known-answer tested against hand-computed values.

5.6 Slashing validation

Monotonicity in miscalibration is verified deterministically at 101 confidence points and by fuzzing (256 runs per direction). Properness is verified numerically by scanning all 101 candidate reports at two distinct true probabilities, $p = 0.30$ and $p = 0.70$, and confirming the expected-loss minimiser equals p in both cases. Four scenarios are then run end-to-end on a local chain (§6.4).

6. Results

All figures below are produced by scripts in the repository. Environment: Windows 11, Python 3.13.5, xgboost 3.1.2, torch 2.9.1+cpu, EZKL 23.0.5, Foundry 1.7.1, laptop CPU, no GPU. Seed 42.

6.1 Calibration

The calibration claims rest on the 10-seed pass. Seeds are pinned in `config.EVAL_SEEDS` and the full list is always run, so a favourable subset cannot be reported selectively.

Head	ECE mean ± std	Brier mean ± std	Accuracy mean ± std
Uncalibrated	0.1853 ± 0.0248	0.2060 ± 0.0191	0.7505 ± 0.0209
Temperature (1 param)	**0.0870 ± 0.0218**	**0.1758 ± 0.0113**	0.7505 ± 0.0209
MLP (321 params)	0.1055 ± 0.0234	0.1908 ± 0.0119	0.7330 ± 0.0199
Control: fitted on TRAIN	*0.2434 ± 0.0212*	—	—

Learned $T = 3.4657 \pm 0.4491$, with $T > 1$ in **every** seed — the base model required softening everywhere, never sharpening. Temperature scaling reduces ECE in **every** seed, and the train-fitted control is worse than not calibrating at all in **every** seed (paired Wilcoxon $p = 0.002$, the minimum attainable at $n = 10$). Accuracy is unchanged by temperature scaling, as it must be: a monotone rescaling of the logit cannot move the decision boundary. Calibration buys a better-priced confidence, not a better classifier, and the table separates the two.

Negative result on head capacity, reported at the strength the data supports. The 321-parameter MLP head does not beat the 1-parameter head. Across the 10 pinned seeds temperature scaling has the lower ECE in **7 of 10** — paired Wilcoxon $W = 7.0$, $p = 0.037$, median difference -0.018. This is a *majority* result, not a uniform one, and it is stated that way deliberately: a single run per head would have supported the stronger and less accurate claim. With 200 calibration points the extra capacity usually fails to generalise, but in 3 of 10 seeds the MLP wins. The reported strength is pinned by `tests/test_multiseed.py::test_temperature_beats_mlp_claim_matches_reported_strength`, so overstating it later would fail the suite. The convenient part of this result is that the better estimator is also the cheaper circuit; that agreement is a measured outcome rather than a design assumption.

Seed-selection audit. Because a single seed can be chosen after the fact, we report where ours falls. Seed 42 ranks 6th of 10 by ECE reduction (48.7% against a 52.8% mean) while carrying the *highest* uncalibrated ECE of any seed — the single-seed figures are therefore pessimistic in absolute terms and middling in relative terms. `SEED = 42` was committed once, before any result existed, and never changed; the repository contains no notebooks.

Seed 42 in detail, since it is the seed the figures, circuits and end-to-end run are generated from:

Head	Params	ECE	MCE	Brier
Uncalibrated	—	0.2164	0.6064	0.2353
Temperature scaling	1	**0.1111**	0.2754	0.1897
MLP	321	0.1436	0.7766	0.2030
Control: temperature fitted on TRAIN	1	*0.2773*	—	—

Learned $T = 3.0121$; base accuracy train 1.0000, test 0.7150. Figure 4 plots the left two columns of this table as reliability curves.

6.2 Explainability

Additivity max absolute error 1.26×10^{-5}; attributions bit-identical across 3 reruns; 5/5 directional sanity checks pass (`installment_rate_pct_income` $+0.621$, `duration_months` $+0.505$, `credit_history` -0.756, `checking_status` -0.943, `savings_status` -0.834).

6.3 Proving and verification

Metric	Temperature (1 param)	MLP (321 params)
Circuit size	2^{15} = 32,768 rows	2^{15} = 32,768 rows
Proving time	**2.09 s**	**2.08 s**
Native verification	0.094 s	0.058 s
Keygen	1.86 s	2.03 s
Proof size	17,447 B	17,566 B
Proving key	132 MiB	132 MiB
Calldata per proof	3,300 B	3,300 B
Tamper-soundness	4/4 rejected	4/4 rejected

Proving cost is invariant to a 16,897× parameter increase. A 10-point sweep from 1 to 16,897 parameters holds `logrows` = 15 at every point. Rows used scale linearly with parameter count (slope 0.98 rows/param, $r = 0.992$) but proving time does not follow it: mean 2.13 s, standard deviation 0.09, regression slope +0.009 s per decade of parameters, Spearman rank correlation -0.188 at $p = 0.603$ — indistinguishable from no relationship. Fixed lookup-argument and column overhead dominate the multiply-accumulate count entirely. This is the answer to RQ3, and it is the reason the design proves a head rather than a classifier: within the circuit's capacity, a larger head is very nearly free, so the binding constraint is capacity, not parameter count.

Scope limit. This holds *within* logrows = 15. The largest head fills 16,511 of 32,768 rows (50.4%), so the sweep does not cross the capacity boundary; where cost steps at logrows 16+ is **unmeasured**. The claim is "flat until the head stops fitting the circuit," not "flat without limit."

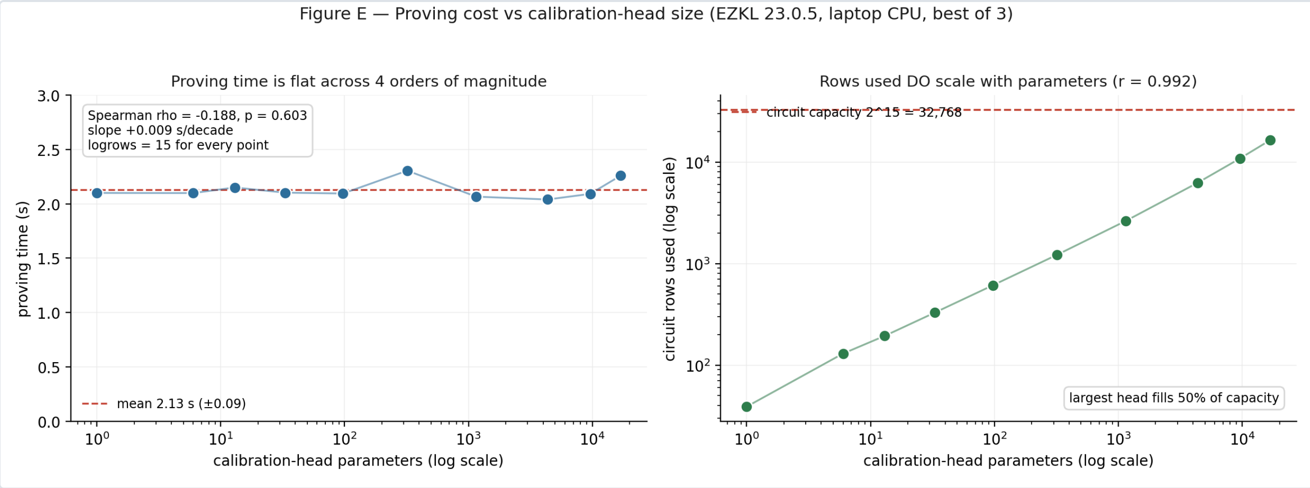

Figure 5 — Proving cost against head size across four orders of magnitude. Rows used scale linearly with parameters; proving time does not follow.

Soundness is tested rather than assumed: an honest proof verifies, while a tampered public output, a flipped byte at the proof head, a flipped byte mid-proof, and a wrong verifying key are each rejected.

6.4 On-chain cost and end-to-end behaviour

Operation	Gas
`Halo2Verifier` deployment (one-off)	2,942,192
`verifyProof` (real EZKL proof)	**684,696**
`attest` (verify + store)	887,376
`openDispute`	134,608
`resolveDispute` incl. slash + payout	97,390
`stake`	45,443
`withdraw`	29,616
`BrierMath.slashAmount` (pure)	543

Verification is ~33× a plain 21,000-gas transfer. The Brier arithmetic is 543 gas — approximately 0.08% of the per-decision cost. **Essentially all cost is halo2 verification, not the mechanism.**

End-to-end, on a local Anvil chain, with a real proof generated per decision:

Scenario	Confidence	Outcome	Slash (% of stake)	Prove
(a) confident + correct	0.9845	upheld	**0.0239%**	2.24 s
(b) confident + wrong	0.9367	overturned	**87.7458%**	2.25 s
(c) uncertain + wrong	0.5012	overturned	**25.1173%**	2.06 s

Ordering holds: (b) > (c) > (a). Confident-and-wrong costs 3,667× confident-and-correct and 3.49× uncertain-and-wrong. Percentages are of the stake remaining when each scenario ran — the three execute sequentially against one shrinking stake, so the percentage column is the comparable one and absolute ETH values across rows are not.

These figures are now enforceable against the operator, which was not true in v0. The earlier exit path — front-running `openDispute()` with an instant `withdraw()` — reduced every slash in this table to zero. Withdrawal is now a two-step request/execute separated by an unbonding delay, and an open dispute freezes execution, so `ThreatModel.t.sol::test_tier2_frontRunDisputeByWithdrawing_isNowBlocked` asserts the attack fails and is the standing regression test for it. What the table still does not survive is a dishonest committee: resolution is a tier-3 action, so the figures describe what the mechanism computes and enforces against the *operator*, not what it enforces against its own resolvers (§7.2, §7.3).

6.5 A deeper base model does not help (Phase A)

An entity-embedding neural network over the same 19 features, and its ensemble with XGBoost, were evaluated on the identical 10-seed protocol. The XGBoost arm reproduces §6.1 exactly, which is the check that the protocol is the same one rather than merely a similar one.

Model	Uncalibrated ECE	Calibrated ECE	Calibrated Brier	Accuracy
XGBoost (baseline)	0.1853 ± 0.0235	0.0870 ± 0.0207	0.1758 ± 0.0107	0.7505 ± 0.0198
Tabular NN	0.2419 ± 0.0189	0.0874 ± 0.0226	0.1843 ± 0.0071	0.7280 ± 0.0187
Ensemble	0.1617 ± 0.0181	0.0834 ± 0.0154	0.1733 ± 0.0091	0.7410 ± 0.0164

Null result. On calibrated ECE the ensemble wins 6 of 10 seeds ($p = 0.6953$) and the NN alone 5 of 10 ($p = 0.9219$) — neither distinguishable from chance. On accuracy the ensemble *loses*, 2 of 10 ($p = 0.1953$). It is not adopted.

The informative part is where the gain goes. The ensemble clearly improves *uncalibrated* ECE (0.1617 against 0.1853), and temperature scaling then absorbs the whole difference. **Ensembling and calibration are substitutes here, not complements.** Since the ensemble costs an entire second model family and calibration costs one parameter, this is an argument for the one-parameter head rather than against sophistication in general.

6.6 Epistemic uncertainty makes calibration worse (Phase C)

5 independently seeded network copies; disagreement is the standard deviation of member probabilities, supplied to the head alongside the margin.

Arm	ECE	Brier	Accuracy
Temperature (baseline, 1 param)	0.0870 ± 0.0207	0.1758 ± 0.0107	0.7505 ± 0.0198
Margin + disagreement	0.1072 ± 0.0324	0.1916 ± 0.0174	0.7295 ± 0.0294
Margin + constant (capacity control)	0.0743 ± 0.0170	0.1757 ± 0.0109	0.7530 ± 0.0165

Negative result, and the control is what makes it legible. The epistemic signal *degrades* calibration: 2W/8L against the baseline, median +0.01611, $p = 0.1309$. But the same two-input head shape fed a *constant zero* in place of the signal beats the baseline — 8W/2L, median -0.01238, $p = 0.0371$.

So the architectural change helps slightly and the epistemic signal then destroys that gain. Without the control this would have read as "ensemble disagreement doesn't help"; with it, the finding is sharper and less comfortable — the signal is **noise that the head overfits** on 200 calibration points. The cost is not close either: 5 model copies at 54 s per seed, roughly ten times the baseline, to make calibration worse.

This is the clearest instance in this work of a claim that sounds obviously true and is not. It is reported at the same prominence as the positive results because a design that only reports its successes is not measuring anything.

6.7 Conformal prediction, and its circuit cost (Phase B)

Empirical coverage across the pinned seeds, at three levels:

Target	Empirical coverage	Min over seeds	Seeds ≥ target	Avg set size
0.80	0.7930 ± 0.0399	0.7300	5/10	1.102
0.90	0.8815 ± 0.0408	0.8200	4/10	1.357
0.95	0.9335 ± 0.0398	0.8550	4/10	1.570

Coverage tracks the target and sits 1–2 points below it at every level. At $n = 10$ the shortfall is within sampling noise, so this is **not** reported as a violated guarantee — nor as confirmation. The likely cause is that the three-way split is stratified, which mildly breaks the exchangeability conformal assumes; on genuinely iid synthetic draws the implementation does meet its target (`tests/test_conformal.py`). Set size is reported alongside coverage because coverage alone is trivially satisfiable: always return both labels and you have 100% coverage and no information.

The circuit gate, measured rather than argued.

Head	logrows	Rows used	Proving time	Proof size
Temperature only	15	39	1.72 s	17,525 B
Temperature + conformal	15	95	1.66 s	17,828 B

It fits inside the established logrows = 15 budget, adding 56 rows of 32,768 and no measurable proving time. The reason is that both comparisons push through the monotone sigmoid into logit space and become comparisons against two constants — so the sigmoid never enters the circuit and the head stays affine. Conformal set construction is therefore **promoted to tier 1**, proved on the same terms as the confidence itself.

Two things this does not buy, stated because they are easy to overstate. The quantile q is not computed in-circuit: that needs a sort over calibration scores and it is a per-deployment constant, so it is fitted off-chain and committed exactly as T is — its provenance is exactly as unproven as T's. And the coverage guarantee is *marginal*: a conformal predictor can hit 90% overall while covering a protected subgroup at 60%, which is the same limitation aggregate ECE has (§7.4).

6.8 Counterfactual evidence (Phase D)

Metric	Value
Rejections examined	40
Counterfactual found	40 (100%)
Mean changes when found	1.075
Immutable-attribute violations	0
Implausible (unobserved) values	0
Mean overlap with SHAP top-5	0.762
Hash reproduces on rerun	True

The plausibility check earned its place on the first run, which flagged 10 of 40 cases: quantile-derived candidate values are interpolated, so the search was proposing loan amounts occurring nowhere in the data. "Reduce the loan to 3,271 DM" is not an actionable instruction when no such loan exists, and candidates are now snapped to observed values. The 0.762 overlap with the SHAP top-5 is a consistency check on the bundle, since the two explanations are committed together and routine disagreement would make the evidence internally inconsistent. The search is greedy, so what is reported is sparsity, not minimality.

6.9 Collusion detection, on synthetic rings only (Phase E)

A two-hop GraphSAGE classifier over the claimant projection of the dispute graph, against a degree-concentration baseline. Intensity is the fraction of a ring member's disputes aimed at the ring's operator.

Ring intensity	GNN AUC	Degree-baseline AUC
1.00	1.000 ± 0.001	0.984 ± 0.009
0.85	0.980 ± 0.013	0.958 ± 0.014
0.70	0.884 ± 0.097	0.865 ± 0.102
0.60	0.856 ± 0.104	0.841 ± 0.104
0.50	0.807 ± 0.040	0.774 ± 0.066

At intensity 1.00 a ring is trivially separable, so reporting only that number would be close to dishonest; the sweep runs to 0.50, where the ring is half camouflaged in ordinary traffic.

This is the weakest evidence in the document, and it now carries the strongest consequences. Labels exist only because the rings were injected. There is no labelled real collusion — the protocol has never run — so no real-world detection performance is claimed or claimable, and the false-positive rate on genuine dispute traffic is unknown.

The detector nonetheless has an on-chain enforcement path (`CollusionOracle`). An enforced flag blocks a claimant from filing disputes and withholds their payout. This is a deliberate decision to give an unvalidated model authority over money, and §7.5 states what it costs.

6.10 Two on-chain surfaces, and where they actually sit

Both additions here are on-chain, and neither is a cryptographic guarantee. The distinction is the whole content of this subsection, because being on-chain is routinely mistaken for being free of trusted parties, which neither of these is.

Operator reputation (Phase F). An exponential moving average of realised Brier score per operator, updated at each resolution from the same *(c, o)* pair that drives the slash — so the score and the penalty cannot diverge. The aggregation is on-chain and auditable: anyone can replay the update rule against the emitted events. But its *inputs* are resolved dispute outcomes, which come from the committee. **The score therefore sits in tier 3 and inherits that trust level exactly**, which `test_tierBoundary_colludingCommitteeCanManufactureReputation` asserts by having two colluding resolvers destroy an honest operator's record with no recourse. What it adds is legibility, not assurance: a number computed by a rule fixed in advance, so an operator's history is visible without trusting anyone's summary of it.

The obvious gaming path is closed by the mechanism itself. An operator farming disputes on decisions it knows are correct still pays $(c-1)^2$ on every sample and cannot reach zero — the irreducible $p(1-p)$ of §3.2, appearing as contract behaviour. Wiring is optional: a pool constructed without a register slashes identically, so reputation can never become load-bearing for the penalty.

Model-version registry (Phase G). A halo2 verifying key covers a *program*, not an *input*: one key verifies every proof from that circuit whatever weights were compiled into it. So `model_version` cannot be inferred from a proof and must be an explicit commitment, or an attestation silently means "some version of this head ran". The registry records a content hash per version, making substitution detectable by anyone.

Its guarantee is content integrity and nothing more, and two tests exist purely to pin that boundary: a deliberately poisoned model registers and verifies perfectly cleanly (`test_boundary_poisonedModelRegistersAndVerifiesCleanly`), and the proof does not bind a version id to the circuit that ran (`test_boundary_anyoneMayReferenceAVersionTheyDidNotRun`). Training integrity is a different and much harder problem and is out of scope. If either test ever needs changing, the claim has been inflated.

6.11 Test suite

110 Solidity tests (Foundry) and 81 Python tests, all passing.

Suite	Tests	What it holds down
`BrierMath.t.sol`	20	Fixed-point properness, monotonicity, fuzzing
`StakePool.t.sol`	24	Staking, disputes, slash cap, payout failure
`ThreatModel.t.sol`	11	The documented weaknesses, as executable evidence
`Unbonding.t.sol`	10	The withdrawal-freeze invariant, fuzzed over timing
`Verifier.t.sol`	4	Real-proof verification and tamper rejection
`ReputationRegister.t.sol`	16	EMA arithmetic and reputation-farming paths
`ReputationIntegration.t.sol`	6	Reputation driven by real resolutions
`ModelRegistry.t.sol`	19	Tamper detection, and the boundary of that claim

Two of these carry unusual weight. `ThreatModel.t.sol` does **not** assert the system is secure — it asserts that each documented weakness is real and reachable, so a failure there means the threat model has drifted from the code and the proposal is wrong. And `tests/test_claim_vocabulary.py` greps this document for language that would attribute properties the system lacks (§7.2); the forbidden list lives in the test rather than in prose, so a later edit cannot quietly upgrade a claim.

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7. Limitations and threat model

These are unresolved trust gaps in the present design. They are not future work.

7.1 Only the calibration head is proved

The zk proof establishes: *given a logit L committed on-chain, the head identified by this verifying key maps L to the attested confidence c.* It establishes nothing about where L came from. The base classifier is not in any circuit. **An operator that fabricates L obtains a proof that verifies**, as demonstrated by `test_tier1_marginIsUnverifiedOperatorSuppliedInput`, in which the chain accepts `type(int256).max` as a margin and still marks the attestation proved. Any description of this system as "zero-knowledge proving the AI decision" is false.

A secondary point: the proof attests the **quantised** computation. At input/param scale 13, the in-circuit fixed-point head differs from the float head by up to 4.2×10^{-4}. The on-chain confidence is the quantised one.

7.2 Dispute resolution is bounded trust, not an absence of trust

`resolveDispute` now requires N of M signatures from a fixed resolver committee, replacing v0's single admin key. A single resolver cannot act alone (`test_tier3_singleResolverCannotResolveAlone`). That is the entire improvement, and it should be read narrowly: **the number of keys an adversary must corrupt goes from 1 to N, and nothing else about the trust model changes.**

Specifically, and each of these is asserted by a passing test:

- N colluding members have exactly the power the single admin had — they can slash a well-calibrated operator to zero (`test_tier3_committeeCanSlashAnHonestOperator`) or shield a miscalibrated one (`test_tier3_committeeCanShieldAMiscalibratedOperator`), which `test_tier3_nOfMCollusionHasSameEffectAsV0Admin` states as the headline case.
- The admin appoints and can replace the committee (`test_tier3_adminCanReplaceTheCommittee`), so the bound applies to the *resolution* step, not to committee *selection*.
- Resolvers have nothing at stake. A resolver who votes dishonestly loses nothing, so there is no economic mechanism holding the tier honest — which is why it is not tier 2.
- There is still no evidentiary standard. Resolvers vote on a bare boolean, and nothing in the contract defines what evidence to consult or penalises consulting none.

`docs/PHASE3B_TRUST_MODEL.md` fixes this vocabulary, and it was written before the code it describes so the claim could not drift upward once the tests went green.

Beneath the implementation gap sits a harder problem, and moving from one voter to N voters does not touch it: for a loan *rejection* the counterfactual is unobservable, because a rejected applicant never demonstrates repayment. "The decision was wrong" has no on-chain referent. This is unsolved, not merely unimplemented.

Beneath the implementation gap sits a harder problem: for a loan *rejection* the counterfactual is unobservable, because a rejected applicant never demonstrates repayment. "The decision was wrong" has no on-chain referent. This is unsolved, not merely unimplemented.

7.3 The unbonding period closes the exit path, and two gaps remain

Withdrawal is a two-step `requestWithdrawal` / `executeWithdrawal` separated by an unbonding delay, with a floor of one hour enforced at construction and seven days used in tests. Any open dispute against an operator freezes execution. `Unbonding.t.sol` fuzzes the invariant over randomised interleavings of request, dispute and execution rather than testing one sequence, and `test_tier2_frontRunDisputeByWithdrawing_isNowBlocked` is the standing regression for the specific v0 exploit — if it ever passes again, tier 2 has regressed.

Two residual gaps survive, and neither is fixed by a longer timelock:

Gap A — exit before any dispute is raised. An operator that requests withdrawal and is not disputed within the unbonding window exits cleanly, and a later dispute lands against an operator with nothing at stake (`test_tier2_residualGap_exitBeforeAnyDisputeIsRaised`). This is a bound on the *dispute window*, not a timelock bug: the unbonding period must exceed the time a claimant realistically needs to notice a bad decision and act. For loan rejections, where the counterfactual may take months to surface, seven days is almost certainly too short, and choosing that parameter honestly is unresolved.

Gap B — tier 2 is enforced downstream of tier 3. The freeze is released by dispute *resolution*, which is a committee action. A dishonest committee can resolve favourably to unfreeze an operator's exit (`test_tier2_residualGap_adminCanUnfreezeByResolvingFavourably`). Tier 2 is therefore enforced against the operator but remains subordinate to tier 3, and closing it requires progress on §7.2 rather than on the timelock.

7.4 Fairness

Attribute 9 (personal status and sex) is excluded from the feature set. This is a floor, not a fairness guarantee: proxy discrimination through correlated features is untested, no disparate-impact analysis was performed, and — most relevant to this mechanism — calibration is measured in aggregate and **not within protected subgroups**. A model can be well calibrated overall while badly miscalibrated for a subgroup, which is precisely the harm this mechanism should detect and currently does not.

The conformal layer of §6.7 does **not** fix this, and it would be easy to imply otherwise. Its coverage guarantee is *marginal*: a conformal predictor can hit 90% coverage overall while covering a protected subgroup at 60%. A stronger uncertainty claim about the population average is still a claim about the population average.

7.5 An unvalidated detector has been given authority over money

This is the sharpest limitation in the system, and unlike the others it is a choice rather than an inheritance.

§6.9 reports detection performance on rings that were injected, so the labels exist by construction. There is no labelled real collusion, the protocol having never run, and the false-positive rate on genuine dispute traffic is therefore unknown. An earlier draft of this document said the detector was not wired to any enforcement action and must not be until that number exists. It is now wired, and the number still does not exist.

What an enforced flag does: blocks the claimant from opening disputes, and diverts any payout they win into quarantine. What it does not do: slash. The operator's penalty is computed and taken identically whether or not the claimant is flagged, because a suspected colluder does not make a bad decision retroactively correct — and because letting a flag reduce a slash would hand operators a motive to get their own claimants flagged.

Four properties bound the damage, each asserted by a test:

1. A flag never causes a slash.
2. An appeal window precedes any effect; a zero-length window is rejected at construction, so a false positive cannot bite in the block it was raised.
3. Quarantined funds are reversible in full when a flag is cleared.
4. Permanent confiscation is a separate, admin-only act, so the only path from a model output to an irreversible loss runs through a person.

None of that makes the detector correct. `CollusionOracle.t.sol` carries three `test_dangerous_*` cases which exist to keep this legible: a false positive silences an honest claimant and thereby *protects* a genuinely bad decision; the reporter key can flag anyone, with nothing on-chain distinguishing a model output from a lie; and flag-plus-forfeit confiscates a legitimate award with recourse only to the admin that took it.

In trust terms this sits below tier 3. The committee at least requires N signatures. This is one key relaying an unvalidated model. Deployments that have not measured the detector's false-positive rate on their own traffic should attach no oracle at all, which `test_poolWithoutOracleIgnoresFlagsEntirely` shows costs nothing else.

7.6 Scope of the empirical claims

Results are from a single dataset (n = 1,000; test split n = 200), one seed, one model family, and one decision class, on a local devnet. Calibration drift over time is not evaluated. Staking parameters are demonstration values with no actuarial basis, and correlated tail risk — one bad model version producing thousands of simultaneous claims — is not modelled.

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8. Proposed work

§7 is a list of reasons the present system should not be trusted. This section is the work that would remove them, ordered by what blocks what. Each item states the deliverable and the evidence that would count as having done it — the point being that these are falsifiable targets, not intentions.

8.1 Enforcement: shipped, with the parameter question still open

W1 — Unbonding period. Done (Phase 3a). Withdrawal is a two-step request/execute separated by an unbonding delay, and an open dispute freezes execution. The acceptance criterion was that `test_tier2_operatorCanFrontRunDisputeByWithdrawing` invert; it now reads `..._isNowBlocked` and asserts the attack fails, with `Unbonding.t.sol` fuzzing the invariant across randomised interleavings rather than one sequence (§7.3).

W1a — Choosing the unbonding period honestly. Open. Residual gap A shows the timelock only helps if it exceeds the time a claimant needs to notice a bad decision and act. For loan rejections the counterfactual may take months, so seven days is almost certainly too short. *Deliverable:* an analysis relating dispute-window length to observed claim latency, and the capital-efficiency cost the delay imposes on an honest operator — the real design tension, still unquantified.

8.2 Making the attested confidence mean something

W2 — Input-logit provenance. Three paths, in decreasing order of guarantee strength and increasing order of practicality:

Path	Guarantee	Cost	Status
Prove the base classifier in-circuit	Cryptographic, end-to-end	Prohibitive for depth-8, 400 trees	Unmeasured; §6.3 suggests capacity, not parameters, is the wall
Commit to feature vector and model weights	Fabrication becomes *detectable*, not *impossible*	Cheap — hashes only	Closest to shippable
Attest the base model in a secure enclave	Hardware trust replaces cryptographic trust	Cheap	Strictly the weakest of the three

The honest ordering matters: only the third is cheap today, and it is the one that gives up the most. *Evidence:* the first path is answered by measuring where proving cost steps past logrows 15 for a real ensemble — the boundary §6.3 explicitly did not cross. That measurement is the immediate next experiment.

W3 — Adjudication beyond bounded trust. Partly shipped (Phase 3b), and the remainder is the hard part. The single admin key is gone: resolution now needs N of M committee signatures, so an adversary must corrupt N keys rather than one (§7.2). That is the whole of the improvement, and `test_tier3_nOfMCollusionHasSameEffectAsV0Admin` exists to stop it being read as more. What remains is everything that would make the tier *accountable* rather than merely wider: a Kleros-style juror pool drawn from a staked set [7], an evidentiary standard, appeals, and resolvers with something at risk. *Evidence:* `test_tier3_committeeCanSlashAnHonestOperator` and `test_tier3_committeeCanShieldAMiscalibratedOperator` must both become unreachable, and `test_tier2_residualGap_adminCanUnfreezeByResolvingFavourably` with them, since gap B is downstream of this tier.

W3 has a hard research problem underneath it, and it should not be presented as an integration task. For a loan *rejection* the counterfactual is unobservable — a rejected applicant never demonstrates repayment — so "the decision was wrong" has no on-chain referent. Any juror pool needs a definition of the adjudicated event before it can vote on one. Candidate framings worth evaluating: restrict disputes to decisions with an observable counterfactual (approvals that defaulted); adjudicate *process* compliance rather than outcome; or score against a delayed, independently sourced outcome oracle. None is obviously right.

8.3 Strengthening the empirical claims

W3b — Bind the committed constants. §6.7 promoted conformal set construction into the circuit, but the quantile q joins T as a fitted constant committed off-chain. Every strengthening of the *proved* computation so far has left the *inputs* to that computation unproven, which is the same shape as the tier-1 gap in W2. Committing (T, q) through the Phase G registry and binding the registry entry into the circuit's public instances would close the constant half of it, and is cheap. It does not touch the logit half.

W4 — Subgroup calibration as the slashed quantity. §7.4 is the sharpest scientific gap: a model can be well calibrated in aggregate and badly miscalibrated for a protected subgroup, which is precisely the harm this mechanism should price and currently cannot see. Slashing on within-group ECE rather than aggregate confidence would target it directly. *Evidence:* a demonstration that the current rule fails to penalise a model constructed to be subgroup-miscalibrated but aggregate-calibrated, and that the revised rule does.

W5 — External validity. The empirical claims rest on one dataset, one model family, and one decision class (§7.6). *Evidence:* replication on a second credit dataset and a second model family, plus a calibration-drift study over time, which is currently not evaluated at all.

W6 — Validate the detector that is already enforcing. The graph analysis recovers injected rings (§6.9) and now has an enforcement path (§7.5), which inverts the usual ordering: the capability shipped before the evidence that would justify it. The appeal path exists; the measurement does not. Required, in order of urgency: a labelled corpus of real disputes; a measured false-positive rate on honest traffic, which is the number that decides whether the oracle should be attached at all; and an appeal reviewer that is not the same admin who can forfeit. Until the second of those exists, the honest deployment posture is to pass `address(0)` and leave the detector reporting into a dashboard.

W7 — Why did the ensemble not help? §6.5 found that ensembling improves uncalibrated ECE and that calibration then absorbs the gain entirely, and §6.6 found that an epistemic-uncertainty signal actively harms calibration on 200 calibration points. Both suggest the binding constraint is the *size of the calibration split*, not model capacity. That is a testable claim: sweep the calibration split size and see whether the two-input head's advantage appears once it has enough data. If it does, the negative results of §6.5–§6.6 are statements about this dataset rather than about the approach, and they should be restated as such.

8.4 Explicitly out of scope

Non-deterministic decision classes (LLM outputs). We flag this as **substantially harder, not a natural extension.** The present design depends on a deterministic decision with a scalar confidence and a binary outcome. Free-form generation has no canonical scalar confidence, no binary ground truth, and no stable notion of "the same decision" across sampling seeds. Applying a proper scoring rule there requires first defining the event space being scored, which is an open research problem and not an engineering task. We would rather name it as out of scope than imply the mechanism generalises for free.

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9. Conclusion

Returning to the four questions of §1.3.

RQ1 is answered affirmatively. The Brier slash has a unique expected-loss minimum at the operator's true subjective probability (§3.2), and the deployed fixed-point Solidity reproduces that minimum numerically at two distinct true probabilities, with monotonicity fuzzed in both directions.

RQ2 is answered affirmatively, with the cost stated plainly. The calibration head proves in 2.09 s on a laptop CPU and verifies on-chain for 684,696 gas — roughly 33× a plain transfer. The mechanism's own arithmetic is 543 gas, about 0.08% of the per-decision cost. Essentially all cost is halo2 verification, not confidence-weighted slashing.

RQ3 is answered, within a stated boundary. Proving cost is flat across a 16,897× parameter increase; capacity, not parameter count, is the binding constraint. The boundary itself — where cost steps past logrows 15 — is unmeasured, and measuring it is the next experiment.

RQ5 is answered negatively, and usefully so. Of five enhancements ablated against the baseline, conformal prediction earns its place — a genuinely stronger uncertainty claim, in-circuit at no measurable cost — while a deeper base model is null and deep-ensemble epistemic uncertainty is actively harmful. The capacity control in §6.6 is what makes the second result interpretable: the architecture helps, the signal destroys the gain. The one-parameter head of §6.1 survives every attempt made here to beat it, which is a stronger endorsement of it than the original single-run result was.

RQ4 is answered negatively, and that is the substantive finding. A proved calibration head is *not* sufficient to make an attestation trustworthy. The proof binds the map from logit to confidence and nothing else: an operator that fabricates the logit obtains a proof that verifies, and the chain accepts it. Beyond that, dispute resolution is an admin-appointed N-of-M committee whose collusion carries exactly the authority a single key did, and while the unbonding period now stops an operator outrunning a dispute, an operator that is simply never disputed inside the window still exits with the stake intact. Each of these is demonstrated by a passing test rather than conceded in prose.

The two on-chain surfaces added since do not change this answer, and were built so as not to appear to. Reputation is auditable aggregation over tier-3 inputs; the registry gives content integrity, not training integrity. Placing each in its real tier was the work, not adding them.

The trajectory is worth stating precisely, because it is the argument for the approach. Two of v0's three named gaps have closed since the first draft, and neither closure required weakening a claim: the exploit tests were *inverted*, and the vocabulary guard in `tests/test_claim_vocabulary.py` prevents the resulting language from drifting upward. What did not move is tier 1 — the input logit is unverified today for exactly the reason it was then — and the ground truth problem beneath tier 3, which more signatures cannot touch.

That invariance is the most durable result here. A better base model, a stronger uncertainty quantifier, a richer evidence bundle, a reputation history and a version registry all now sit on top of the same unproven input. Sophistication above the trust boundary does not move the trust boundary, and this work is a fairly thorough demonstration of that.

The two claims should therefore be separated cleanly. The **mechanism-design claim** — that confidence-weighted slashing on a strictly proper scoring rule makes honest confidence reporting loss-minimising, and that it is cheap to implement and verify — is supported by the evidence presented here. The **systems claim** — that this constitutes a trust-minimised accountability protocol — is not supported. §8 is a plan for closing that distance rather than an assertion that it is nearly closed, and it now distinguishes what has shipped from what remains. The most useful result of the prototype may be the negative one: it establishes precisely which parts of "verifiable AI accountability" the cryptography actually buys, and which parts remain a matter of whom you trust.

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Appendix A — Reproducing the results

Every number in this document is produced by a script in the repository; none is transcribed by hand. The sequence below regenerates all of them from a clean checkout. `10_train_calibrate.py` carries an ECE gate that fails loudly if calibration does not reduce error, so a silently broken run cannot produce a quiet result.

```
pip install -r requirements.txt
bash scripts/00_setup_tools.sh          # EZKL CLI v23.0.5

python scripts/10_train_calibrate.py    # §6.1  (ECE gate; fails loudly if not reduced)
python scripts/20_explain.py            # §6.2
python scripts/30_export_onnx.py        # ONNX export, fidelity-checked
python scripts/31_zk_prove.py           # §6.3  circuits, proofs, soundness
cd contracts && forge test              # §6.11  53 tests incl. ThreatModel.t.sol
python -m pytest tests/ -q              # §6.11  29 tests

anvil &                                  # §6.4  end-to-end
forge script script/Deploy.s.sol:Deploy --rpc-url http://127.0.0.1:8545 --broadcast
python scripts/40_demo_e2e.py
python scripts/90_render_results.py     # regenerate RESULTS.md
python scripts/91_figure_d_reliability.py   # Figure 4
python scripts/13_phaseA_ablation.py     # §6.5  ensemble ablation
python scripts/14_phaseB_conformal.py    # §6.7  conformal coverage
python scripts/15_phaseB_circuit_gate.py # §6.7  circuit feasibility gate
python scripts/16_phaseC_deep_ensemble.py # §6.6  epistemic uncertainty
python scripts/17_phaseD_counterfactual.py # §6.8  counterfactual evidence
python scripts/18_phaseE_collusion.py    # §6.9 collusion detection
python scripts/19_phaseH_ablation_report.py  # regenerate ABLATION.md

python scripts/92_render_svg.py             # Figures 1 and 3
python scripts/93_figure_e_sweep.py          # Figure 5
```